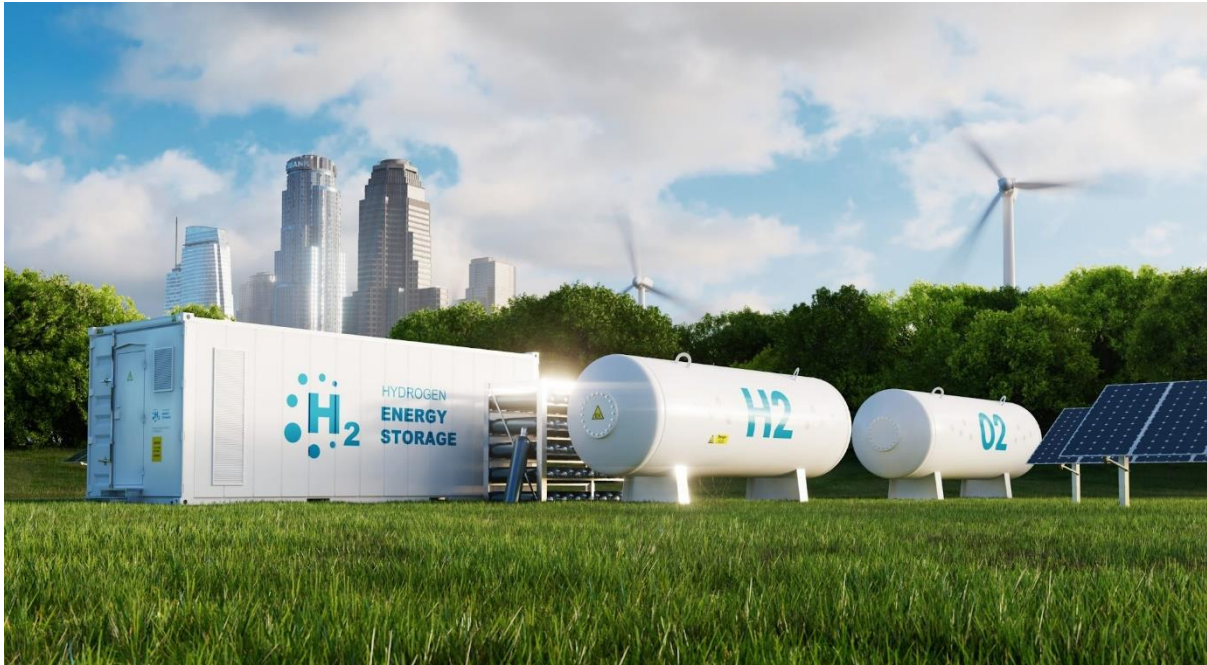


MJ2411 Renewable Energy Technologies
Energy Storage Technologies, Group 16, Project D

‘Power to X’ (P2X)



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Introduction

'Power to X' - or simply 'P2X' - is an umbrella term for a number of conversion, storage, and reconversion pathways that use surplus electricity produced from renewable energy sources, typically solar and wind. The "X" refers to the type of energy carrier into which the surplus electricity is being converted, and are generally either gases or liquids ([Turbomachinery, 2020](#)). The most popular energy carrier surplus energy is thought to be converted into is hydrogen, which then, either be used directly, or be further transformed into hydrogen derivatives such as methanol, methane, or ammonia ([IEA-AMF, 2023](#)). The process through which hydrogen is intended to be produced from renewable energy is called 'electrolysis', which is the process of using electricity to split water into hydrogen and oxygen. For the electrolysis to take place, units called 'electrolyzers' are needed. Electrolysers can in turn range in size from small appliance-size equipment that is well-suited for small-scale distributed hydrogen production, to large-scale central production facilities that could be tied directly to renewable energy sources for electricity production ([DOE, 2023](#)).

The rationale behind the usage of P2X through 'green hydrogen' production, which is the term for hydrogen being produced from electrolysis using renewable energy ([OECD, 2022](#)), is multifaceted. The European "Green Deal," for example, is a set of policy initiatives from the European Commission with the aim of making Europe climate-neutral by 2050 ([EU, 2023](#)). It includes the phase-out of coal, a substantial increase in renewable generation, and ambitious targets to cut greenhouse gas (GHG) emissions by 2030 and decrease air pollution. Wind and solar energy represent the main pillars of these plans, but hydrogen has a complementary role. It is seen as one way to help enable large-scale integration of renewables into the power generation network, as a means of distributing energy across sectors and regions, and as a storage buffer to increase system resilience ([Orsted, 2023](#)). Wind and solar require large-scale energy storage to compensate for short-term and seasonal imbalances, and due to the orders of magnitude involved, hydrogen has come to be seen as a key enabler for higher utilization of renewable energy, as well as for decarbonizing hard to abate industries through its derivatives.

Hydrogen can also be produced in numerous other ways than through electrolysis using renewable energy, including fossil fuels such as coke and natural gas, biomass, crops, and more traditional non-renewable sources for electricity generation. For any other type of hydrogen production than through electrolysis using renewable energy, fossil fuels will in some way be consumed and add to emissions and cannot, thus, be called 'green' or 'clean'.

In this paper, the aim is to address some of the main use cases for P2X and provide examples of how it is utilized in today's energy system, or what the future ambitions are. This will be achieved through answering the following questions:

- Q1.** "Give three conceptualization paths of power to X considering full conversion, transportation, storage, and utilization in an energy chain."
- Q2.** "Give three real-life examples of power to X with concrete calculation examples on energy efficiency, energy losses, and economic aspects, among others."
- Q3.** "Reflect on the utility and the limitations of Power to X concept. What are the drawbacks? Is it always the best option to have intermediate energy? conversions (motivate your results)?"

Methodology

For this paper, lecture slides and videos for the 'Energy Storage' module in the course 'MJ2411 Renewable Energy Technologies' have been used as sources for information as well as guidance for further online research. As online sources for information, industry, government, and academic sites and papers have all been reviewed and used when considered appropriate after having been critically reviewed

and discussed. Though the ambition with this paper is to provide a comprehensive introduction to the topic of P2X and description of use cases, the extent of the assignment did at the same time not allow for a complete description of all existing P2X themes currently circulating in the literature.

Results

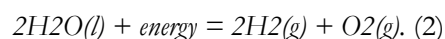
Q1. The main conversion process from electrical energy to chemical energy for the three P2X routes covered in this section is electrolysis, which uses surplus energy when the power grid is underutilized to generate hydrogen and oxygen from inexpensive electricity. The energy conversion efficiency for electrolyzers is, in general, around 75%, but it depends on what type of electrolysis cell that is being used. One Australian company named Hysata claims that their electrolyzer has 95% efficiency ([New atlas, 2022](#)). There are three types of electrolyzers; AEL (alkaline water electrolysis), PEMEL (Proton exchange membrane electrolysis), and SOEL (Solid oxide electrolyzer cell). AEL electrolyzers are currently the most mature and established technology, as well as the most economical, but the others are also used and under development ([Karim, 2018](#)).

Path 1 - Power to Methane: The hydrogen that is produced from the electrolysis process is combined with carbon dioxide, and the reaction is called the 'Sabatier reaction'. The chemical reaction is defined as equation 1, and the reaction creates methane plus water.



Carbon dioxide can be supplied from external sources as either liquid or gas, derived from either biomass, or combustion from power plants. However an american company called Equatic has developed a process that removes carbon dioxide from the ocean by using sea water and an electrolyzer. The captured CO₂ is mineralized and transformed into rocks. ([Energynews, 2023](#)) The methane can be stored in tanks for later usage, either with existing transportation infrastructure using trucks with tanks or directly pumped out in existing gas pipes for natural gas. Methane is most commonly stored as gas, but is liquefied for use in the transportation sector as combustion fuel in truck engines and ship engines. One research study determined that it would be possible to produce 1200 TWh of renewable methane from planned and announced electrolyzer facilities in 2050, sufficient to meet EU gas demand for several months. ([Store and go, 2020](#))

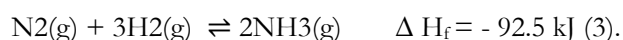
Path 2 - Power to Hydrogen: In this path, a power to hydrogen use case is described. The chemical reaction for the electrolysis is defined as equation 2.



The minimum energy required for this reaction is 39.4 kWh/kgH₂ at standard conditions, but due to energy losses in electrolyzers, more energy is required in real scenarios. According to data from the IEA, by 2030, hydrogen energy generated by electrolysis will be cheaper than natural gas. ([Sean O'Neil, 2019](#)) Industrial electrolyzers have historically been designed to operate continuously, mostly on predictable supply of fossil fuels. Electrolyzers for P2X will have to be capable of operating more dynamically to accommodate the availability of electricity, price changes, and demands. When several electrolysis cells are combined, they are called a stack. An electrolyser's capacity depends on how many stacks it consists of. Economies of scale effects are seen up to 0.5 MW and 2 MW, but beyond that, the costs are linear ([Linus Engstam, 2021, 11](#)). The efficiency of an electrolyzer increases with decreasing load, which makes it suitable for dynamic applications like P2X. After the electrolysis, the hydrogen produced may undergo a purification procedure to remove moisture and impurities. After that, the hydrogen can be put to immediate use, or be stored for later use. When storing hydrogen it must be cooled to a liquid phase with the use of external heat if it is to be kept in liquid form. The mode of storage, either as gas or liquified, determines the requirements for how the hydrogen can be transported. Examples include liquid hydrogen supplied in low-temperature insulated containers, which has a higher energy density and bulk density than

when it is stored as a gas, but is more expensive and increases the complexity, and gaseous hydrogen transported via pressurized long-distance tube trailers or pipelines. High-pressure gaseous hydrogen storage, low-temperature liquid hydrogen storage, high-pressure and low-temperature liquid hydrogen storage, metal-hydride hydrogen storage, organic liquid hydrogen storage, etc. are some examples of the various forms of hydrogen storage. For final and direct usage of hydrogen, there are applications under larger commercial development in the transportation sector. There, hydrogen used by 'Fuel Cell Electric Vehicles' (FCEV) and hydrogen-fueled vehicles blended with natural gas are under larger commercial development. Hydrogen can also be used in gas turbines or fuel cells to generate electricity.

Path 3 - Power to Ammonia: Ammonia (NH₃) is the second largest synthetic inorganic commodity produced worldwide with 80% of the production used by the fertilizer industry. It can easily be stored as a liquid in atmospheric pressure by cooling it to -33 °C, or by pressurizing it at 9 bar at room temperature. The volumetric energy density of liquid Ammonia is 12.679MJ/m³ at 1 bar. Ammonia is a derivative to hydrogen and by producing it from green hydrogen, greenhouse gas emissions could be further reduced. The combined application of water electrolysis and the 'Haber-Bosch' process is called power-to-ammonia (P2A) technology. The reaction to produce Ammonia is exothermic giving off 92 KJ of energy per 2 mol of NH₃ produced. The reaction is carried out at pressures ranging from 100 to 250 bar, and temperatures ranging from 350 °C–550 °C. The chemical reaction is defined as equation 3.



Ammonia has a lot of use cases, and to mention a few; it can be directly used as a refrigerant, to produce fertilizers, for direct combustion, or, can be fed to Fuel cells as hydrogen after ammonia cracking. In contrast to hydrogen in its pure form, there is a large existing infrastructure in place for the transportation and storage of ammonia, and the compound is regularly transported in carbon-steel pipelines, rail cars, trucks or ships ([Jussi et al,2018](#)).

Q2. Path 1 - Power to Methane: Starting with looking at real-life examples and related energy efficiency and economic aspects for power to methane, one specific study looked at the total lifetime cost for a gas plant producing methane ([Janke et al, 2022](#)). In the study, a model for calculating the gas production cost (GPC) (see appendix A) was developed. The model considered the capital expenditures, yearly operational expenditures, yearly variable operational expenditures due to the power consumed, yearly synthetic production in kilos, and 'k' as the discount rate, estimated at 6,5%, for calculating the present value of the plant's expected lifetime costs. Based on this model, it was estimated that the CAPEX cost per kilo-watt installed varied from 1659-3179 €/kW depending on power availability and fuel demand scenarios. And that the gas production cost for methane varies from 162.34-234.42 €/MWh. The price for gas plants are high today but a decreasing trend due to size and experience is visible ([Sayed M, 2017](#)). For further values from the study, see appendix A.

Path 2 - Power to Hydrogen: Considering the transportation sector, the usage of FCEVs could have a huge impact on multiple of the United Nations' (UN's) sustainable development goals (SDGs), to mention a few, SDG 7 - "Affordable and Clean Energy", SDG 9 - "Industry, Innovation and Infrastructure", SDG 11 - "Sustainable Cities and Communities", SDG 12 - "Responsible Consumption and Production", SDG 13 - "Climate Action", SDG 14- "Life below Water" and SDG 15 - "Life on Land" ([Sidhartha H, 2023](#)).

As for the use case of power to methane, efficiency and economical aspects related to the power to hydrogen use case in a fuel cell are presented below, with all assumptions for the calculations provided in appendix B.

Starting with the economic aspects, filling a 5kg hydrogen tank in the car would cost \$3.13/kg * 5kg = \$15.65. The vehicle's yearly maintenance expenses are computed as 2% of \$62,000, or \$1,240 per year. If the car goes 120 km per kilogram of hydrogen and you drive 15,000 km each year (assume), you'll require

15,000 km / 120 km/kg = 125 kg of hydrogen per year. The yearly fuel cost would be \$15.65 per kilogram * 125 kg = ~\$1,960. Annual upkeep (\$1,240) and yearly fuel expenditures (\$1,960) would equal \$1,240 + \$1,960 = \$3,200 per year in total operating costs. The vehicle would convert 5 kg * 33.3 kWh/kg (LHV) * 0.60 = 100 kWh of energy into propulsion with a 60% efficient fuel cell and 5 kg of hydrogen. If the car travels 15,000 kilometers per year and consumes 100 kWh of energy, it will reach 15,000 kilometers / 100 kWh = 150 km per kWh of energy. We can estimate the operating cost per kilometer by dividing the overall running cost (\$3,200) by the yearly mileage (15,000 km): \$3,200 / 15,000 km = \$0.2133 per km.

The following energy losses may occur in an FCEV: Electrolysis Energy Losses - 20% to 30% of power is lost during electrolysis ([Flux Power, 2021](#)); Compression and Storage Losses - Because compression efficiency ranges from 90% to 95%, some energy is wasted as heat during compression. Storage losses can also occur as a result of H₂ leakage or diffusion through materials; Transportation and distribution losses might occur as a result of friction and heat dissipation inside pipes or high-pressure tanks. The total efficiency of FCEV is lower than that of direct power battery storage when all of the aforementioned losses are included. This means that only 30% to 40% of the electrical energy utilized to produce hydrogen is efficiently transferred back into electricity for vehicle propulsion.

Path 3 - Power to Ammonia: According to a study by IRENA Green Ammonia production costs have already become competitive compared to fossil-fed ammonia production, with the current capital expenditure costs being estimated at USD 720-1 400 per tonne of produced ammonia, and they are likely to fall to USD 310-610 per tonne by 2050 primarily driven by decreasing costs for renewable energy ([IRENA, 2022](#)). This is simply the case because electricity is the dominant operational cost factor for large-scale renewable ammonia production, typically accounting for more than half of the lifetime for producing renewable ammonia ([IRENA, 2022](#)). For this reason, unlike fossil-based ammonia plants, many of the renewable ammonia plants currently being under development include the electricity generating capacity within the proposed investment, effectively shifting electricity input from an operational cost to a capital cost ([IRENA, 2022](#)).

An example plant where on-site electricity generation exists, is a plant in Puertollano, Spain (by Fertiberia), for a capacity of 200 Kt/y, Capex cost is USD 2124 Million (in terms of CAPEX USD/t/y - USD 10 620). Some examples where energy generation is not included in CAPEX (where energy is consumed using renewable sources like wind, solar, and hydro) - an Example plant is Port Bonython, Australia run by ThyssenKrupp for 40 T/Y plant Capex stands at USD 186 Million (In terms of CAPEX USD/t/y - USD 4 660). Modern renewable ammonia synthesis from low-temperature has an energy conversion efficiency of 50 % and high-temperature electrolysis has a conversion efficiency of around 60-70%([IRENA, 2022](#)).

Q3. The current weakness of P2X is the low competitiveness of the technology. Because of the high investment cost of electrolyzers and the higher price of electricity compared to other energy carriers such as gas, P2X currently lacks cost efficiency compared to other hydrogen production processes. As a reference, less than 1% of global hydrogen production in 2022 came from renewable energy sources, with cost competitiveness being the significant factor ([IRENA, 2022](#)).

In addition, P2X is a power-intensive technology, which leads to low energy efficiency with currently available technologies and the immaturity and short lifetime of the electrolysis. Even if some electrolysis technologies, such as alkaline, have a high 'Technology Readiness Level' (TRL), the lifetime of electrolysis units might be relatively short compared to traditional plant technologies, and when it comes to linking to electrofuel production, or 'sector coupling', most of the technologies have not been demonstrated on a larger scale and can thus still be defined as immature. Many energy conversions in the P2X energy chain are not good for overall energy efficiency, as energy for conversion equipment and devices are needed, and heat losses are generated.

Conclusion

Though some aspects of the current state still are not fully explored and tested in the P2X value chain, there are understandably several upsides to the successful implementation of larger-scale green hydrogen sight, that would help meet climate targets. To mention a few just by highlighting the P2X provided in this report, successful implementation and technology could help decarbonize parts of the transport sector that are not suitable for direct electrification, such as heavy-duty road transport, aviation and shipping. The conversion of CO₂ into methanol is also an attractive direction because of the high-energy density of liquid fuels that can be readily stored, transported, and utilized for chemical manufacturing. The list of use cases contributing to the nations net zero targets are plentiful, and hopefully, as many as possible can come to fruition.

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APPENDIX

A. Power to Methane Gas Production Cost (GPC)

$$GPC = \frac{CAPEX + \sum_{y=0}^n OPEX_y + Power_y}{\frac{(1+k)^y}{\sum_{y=0}^n \frac{SMP_y}{(1+k)^y}}}$$

Summary of the optimal plant configurations for the power availability and fuel demand scenarios.

Power availability scenarios	Fuel demand scenarios	PEM electrolyser [kW]	CO ₂ storage [kg]	BM reactor [kg CH ₄ /h]	CH ₄ storage [kg]	Electrolyser FLH [h/year]	Methanation FLH [h/year]	CAPEX [€/kW _{el}]	Price [€/MWh]	GPC [€/MWh]
Constrained	Low	3000	8000	110	5000	3905	3842	2309	51.39	234.42
	Mid	6000	19,000	220	7000	3902	3847	1895	51.35	207.14
	High	9000	30,000	330	10,500	3900	3828	1659	51.34	194.81
Unconstrained	Low	1500	8000	55	3500	7703	7685	3179	54.47	188.31
	Mid	3000	21,000	110	7000	7717	7698	2467	54.48	170.78
	High	4500	33,000	165	11,000	7720	7693	2164	54.49	162.34

Note: GPC refers to the levelised cost of synthetic CH₄ production.

B. Power to Hydrogen and Fuel Cell Usage Assumptions

Cost of Green Hydrogen Production:	The cost of green hydrogen production through water electrolysis powered by renewable energy ranges between \$2.94/kg and \$3.32/kg ~ \$3.13/kg on average (Qusay Hassan et al., 2023)
Fuel Cell Efficiency:	Assume fuel cell efficiency of 60% (U.S. Department of Energy, 2015, 1)
Hydrogen Storage:	Assume a typical hydrogen storage capacity of 5 kg in the vehicle's tank (U.S. Department of Energy, 2017, 1)
Cost of Fuel Cell Vehicle:	Assume the cost of the hydrogen fuel cell vehicle is \$62,000 (Hyundai Nexo)
Maintenance Costs:	Include annual maintenance costs of 2% of the vehicle's purchase price.
Cost of Hydrogen Refueling:	Assume the cost of hydrogen refueling is \$16 per kg.
Vehicle Mileage:	Assume the vehicle travels 120 km per kg of hydrogen. (Hyundai,2018)